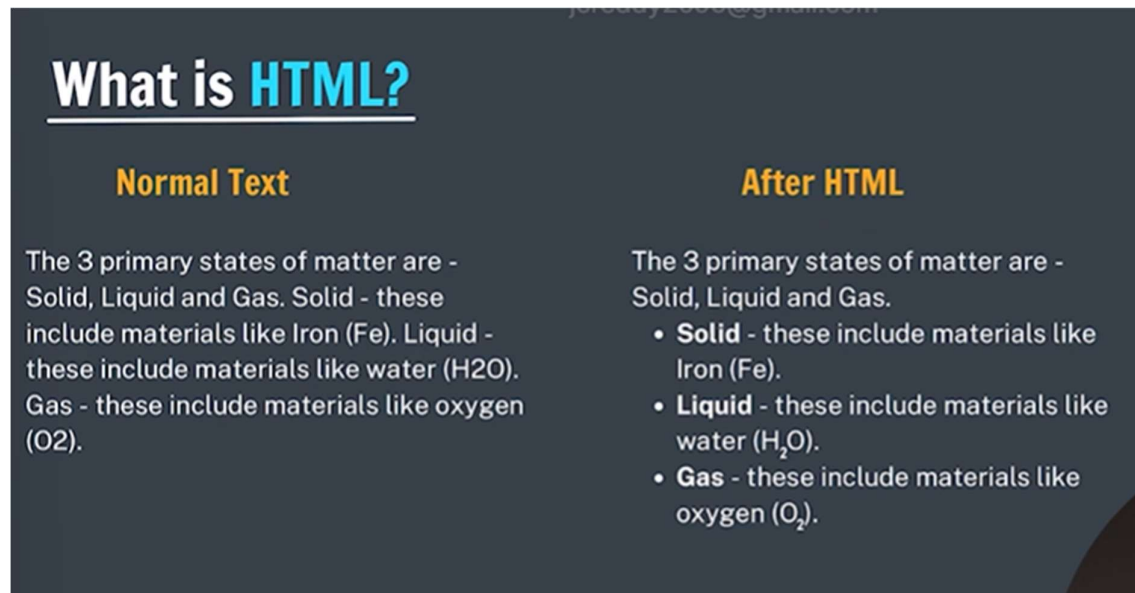


-->html-hypertext markup language

->it do the structure and formatting of the webpage.

->the difference between the normal text and after doing the html we can add a perfect structure to represent it and have a some format

-->



→html elements:

->it is standard elements that browser recognises

Paragraph <p> </p>

Heading <h1> </h1>-----<h6> </h6>

Image

->the component used to design the structure of websites are called html tags.

-><p>→it is opening tag

</p>→it is closing tag

<p> this is paragraph<p/>→it is a paragraph tag

this is paragraph→content

→hello world in html

`<p>helloworld</p>`

->paragraph tag will use the next line if we use multiple paragraph tags

->bold tag `<b> word</b>`

→nesting tags:the tags we written inside another tag

→heading elements:the `<h1>` to `<h6>` html elements represent six levels of section headings `<h1>` is the highest section level and `<h6>` is the lowest.

->html boilerplate

This is the standard format or skeleton of writing html code

`<DOCTYPE html>`->it tells the version of html

`<html>`→root tag

Hierarchial arrangement in html is html ->head ->title

`<head> </head>`→in this we use for to link with the css and js and styling and also for giving extra information about the website int his meta information will be present.

`<title> </title>` →it is for naming the website like it will not present in the web page it is int the title bar.

`<body> </body>` →whatever we are seeing in the webpages all is written in the body tag.

→indentation ->proper spacing

→ lists in html

Unordered list

<ul> → unordered list

<li>bread</li> → list item

<li>jam</li>

<li>vinayaka</li>

</ul>

→ ordered lists: in this we get the order like a,b,c,d or 1,2,3, for like this



<ol>

<li>bread</li>

<li>om</li>

</ol> in this it will print as 1,2,3 normally but if we want a,b,c,d or A,B,C,D..... for this we need to write <ol type="A"> or <ol type = "a">

HTML Attributes: attributes are used to add more information to the tag

For ex:

`<html lang="en">` lang is attribute and en is the value

→ anchor element

Used to add links to your page. By this we can move from one page to another page by clicking it .

Syntax:

`<a href="https://google.com">google</a>`

Here href is the attribute and href: hyper text reference

There are two types of links we use

Absolute : directly we paste the website links from the chrome

Relative: we use this relating one file to another file in our laptop .

→ image element

Used to add image to your page

``

For image tag there is only single tag

In this we can get image by getting the source link if it is not in the same folder if it is in the same folder then we just give the name of the picture

→ Br tag

Used to add next line (line breaks) to your page

Syntax

`<br>`

Bold, italic & underline tags

`<b> bold </b>`

`<i>italic</i>`

`<u> underline</u>`

→ comments in html

<!-- comments --->

Comments are ignored by the compiler it is not printed or any I just for understanding by the user.

To comment out a selected line we use this

Ctrl+forward slash(/)

→html is not case sensitive

Case sensitive is there means a and A both are not same.

If there is no case sensitive then both are same.

→java script is a case sensitive language and also java,c++ also same.

→difference between inline v/s block

Block elements:

Takes up the full width available

It starts from new line

Inline elements

Takes up only necessary width

Don't start from the new line

### Inline v/s Block

| Block Elements                                  | Inline Elements               |
|-------------------------------------------------|-------------------------------|
| Takes up the full-width available (whole block) | Takes up only necessary width |
| Start from new line                             | Don't start from new line     |

Am I inline or block

Header it is block element

->paragraph element it is block element

->anchor tag is inline element

->image element is inline element

→div element

Div is a container used to hold other html elements or group elements together

It is block element

→span element

->Span is also a generic container used to hold other html elements or group elements together

->it is an inline element.

→hr tag

->horizontal rule element

<hr/>

->this tag will give a horizontal line



Sub&sup tag

->subscript and superscript

*subscript* *superscript*

## Sub & Sup Tag

$a^2 = b^2 + c^2$  ← Superscript      `a <sup> 2 </sup>`

$H_2O$  ← Subscript      `H <sub> 2 </sub> O`

```
> Classroom.html > html > body > h1 > sub > u
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>HTML - Level 2</title>
8    </head>
9    <body>
10     <h1>a<sup>2</sup></h1>
11     <h1>
12       H<sub><u>2</u></sub>
13       >O
14     </h1>
15   </body>
16 </html>
17
```

## Semanatic markup

->it is the markup that relates to the meaning of content.

->semanatic means by seeing the tag we can understand what it means

ex:header,paragraph,img

->non-semanatic:by seeing the tag we cannot understand what the tag  
what work it will do

Ex:div,span

→why we use semantics

->by seeing it will be meaninigful

->seo friendly seo:search engine optimization

->readable and user experience

Semantics markup tags under it

<header> </header>

<main> </main>

<footer> </footer>

<nav> </nav> ->it is for the navigation purpose

<article></article>

<section> </section> it works like the div tag in it related content will be  
there

<aside> </aside> it consists of related links/contents



→html entities

## HTML Entities

- An HTML entity is a piece of text ("string") that begins with an ampersand (&) and ends with a semicolon (;).
- used to display **reserved characters** (which would otherwise be interpreted as HTML code), and **invisible characters** (like non-breaking spaces).
- can also use in place of characters that are difficult to type with a standard keyboard.
- browser interprets them & renders correct character

->for space we write &nbsp;

->In html < will directly not visible so we want to write it as &lt;

->for > &gt;

| Character | Entity | Note                                                                           |
|-----------|--------|--------------------------------------------------------------------------------|
| &         | &amp;  | Interpreted as the beginning of an entity or character reference.              |
| <         | &lt;   | Interpreted as the beginning of a <a href="#">tag</a>                          |
| >         | &gt;   | Interpreted as the ending of a <a href="#">tag</a>                             |
| "         | &quot; | Interpreted as the beginning and end of an <a href="#">attribute</a> 's value. |

→ understanding html 5

## Understanding **HTML 5**

The term HTML5 is essentially a buzzword that refers to a set of modern web technologies. This includes the [HTML](#) Living Standard, along with [JavaScript APIs](#) to enhance storage, multimedia, and hardware access.

① new features/  
upgrades  
in  
HTML + other things/features  
not part of HTML

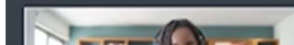
## How **HTML** works?

**HTML Standard** → living standard

HTML Standard is a document that tells to the browser how HTML should work.

html code  
↓  
browser → HTML standard  
↓  
render web page

good read for more details



→ video tag:

<h2>Example of Video Tag</h2>

<!-- Video with controls -->

<video src="sample.mp4" width="400" controls></video>

## →tables in html

Generally in tables will consist of rows and columns

<tr> ->used to display table row. Here it will create a row a single row

<td> -> used to display table data . it is the data which we written inside the row

<th> ->used to display table header ->it is for writing the main headings in the table

## →semnatics in tables

<thead> to wrap the table header.

<tbody> to wrap table body

<tfoot> to wrap table footer

## →Colspan & rowspan attributes

Used to create cells which spans over multiple rows or columns.

## →forms in html

Forms are used to collect data from the user.

Syntax:

<form>

Form content

</form>

## →action attribute

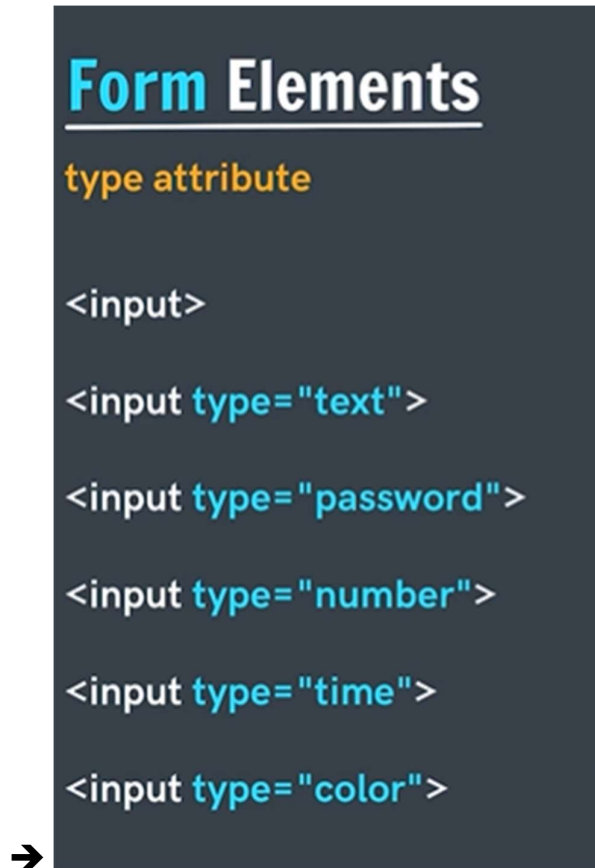
Action attribute is used to define what action needs to be formed when a form is submitted or where the form data should be sent.

<form action="/action.php">

→ form elements

Input

- Used to create multiple form controls
- There are multiple types of inputs that can be created using type attribute.



→ place holder

->it is a attribute

-> <input type="text" placeholder="enter name">

→label

- Label elements represents a caption for an item in a user interface

→ <label>

Enter username:

<input type="text" placeholder="username">

</label>

## Form Elements

### Label

- Label element represents a caption for an item in a user interface

<label >

Enter your username :

<input type="text" placeholder="username">

</label>

## Form Elements

### Label Classical Approach

*id → unique name  
element*

jcreddy2096@gn

<label for="username" Enter your username : </label>

<input type="text" id="username" placeholder="username">

→button element

<button>submit</button>

## Button Element

**type** attribute

**<button type="submit">** submit **</button>**

**<button type="button">** do something **</button>**

**<button type="reset">** do something **</button>**

## Input Element

**Button using input**

**<input type="submit" >**

**<input type="submit" value="click me">**

```

oom.html > html > body > form > input
<head>
  <meta charset="UTF-8" />
  <meta http-equiv="X-UA-Compatible" content="IE=edge" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>HTML - Level 3</title>
</head>
<body>
  <button type="submit">hello</button>
  <form action="/server">
    <div>
      <label for="username">Username : </label>
      <input type="text" placeholder="enter username" id="username" />
    </div>
    <div>
      <label for="password">Password : </label>
      <input type="password" placeholder="enter password" id="password" />
    </div>
    <div>
      <label for="age">Enter Age : </label>
      <input type="number" placeholder="age" id="age" />
    </div>
    <button>Submit</button>
    <button type="button">hello</button>
    <button type="reset">reset</button>
    <br />
    <input type="button" value="click me" />
    <input type="submit" value="click me b2"/>
  </form>

```

--

→Name attribute:

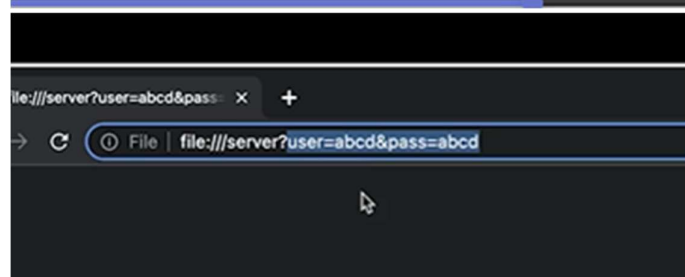
jcredy2096@gmail

## Name Attribute

Name of the form control. Submitted with the form as part of a name/value pair

```
<input type="text" placeholder="enter name" id="username" name="username"/>
```

```
Classroom.html > html > body > form > div > input#password
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>HTML - Level 3</title>
  </head>
  <body>
    <form action="/server">
      <div>
        <label for="username">Username : </label>
        <input
          type="text"
          placeholder="enter username"
          id="username"
          name="user"
        />
      </div>
      <div>
        <label for="password">Password : </label>
        <input type="password" placeholder="enter password" id="password" name="pass"/>
      </div>
      <button>submit</button>
    </form>
  </body>
</html>
```





→checkbox:

reddy2096@gmail.com

| am 18+ ☐

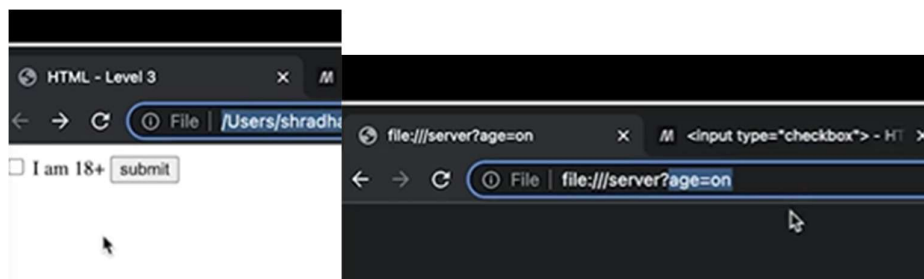
# Input Element

## Checkbox

```
<input type="checkbox" name="age" id="age" checked />
<label for="age"> I am 18+ </label>
```

```
<> Classroom.html x jcreddy2096@gmail.com
<> Classroom.html > html > body > form > button
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7     <title>HTML - Level 3</title>
8   </head>
9   <body>
10    <form action="/server">
11      <input type="checkbox" id="age" name="age" checked />
12      <label for="age"> I am 18+ </label>
13      <button>submit</button>
14    </form>
15  </body>
16 </html>
17
```

If we put tick mark in checkbox then it passes as on



→ Radio Button

## Input Element

### Radio Button

name = value  
↓  
fruit

- ☐ apple
- ☐ mango
- ☐ banana

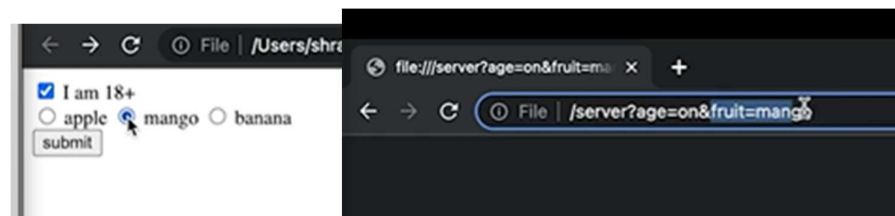
```
<input type="radio" name="fruit" id="apple" value="apple" />
```

```
<label for="apple"> Apple </label>
```

jcreddy2096@gmail.com

classroom.html > html > body > form > div > input#banana

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>HTML - Level 3</title>
  </head>
  <body>
    <form action="/server">
      <input type="checkbox" id="age" name="age" checked />
      <label for="age"> I am 18+ </label>
      <div>
        <input type="radio" name="fruit" id="apple" value="apple" />
        <label for="apple">apple</label>
        <input type="radio" name="fruit" id="mango" value="mango" />
        <label for="mango">mango</label>
        <input type="radio" name="fruit" id="banana" value="banana" />
        <label for="banana">banana</label>
      </div>
      <button>submit</button>
    </form>
  </body>
</html>
```



- If we didn't mention value attribute then fruit = on will be shown if we mention value then it will be shown like this
- For the name attribute we put same for all because we want any one of them if we put unique then we will select all.



File | /Users/shradhakapra/WebClassroom/Classroom.html

☒ I am 18+  
☐ apple ☒ mango ☐ banana

Choose your subjects: ☒ math ☒ science ☒ english

→

```
ssroom.html > html > body > form > div > input#math
<body>
  <form action="/server">
    <input type="checkbox" id="age" name="age" checked />
    <label for="age"> I am 18+ </label>
    <div>
      <input type="radio" name="fruit" id="apple" value="apple" />
      <label for="apple">apple</label>
      <input type="radio" name="fruit" id="mango" value="mango" />
      <label for="mango">mango</label>
      <input type="radio" name="fruit" id="banana" value="banana" />
      <label for="banana">banana</label>
    </div>
    <br />
    <div>
      Choose your subjects:
      <input type="checkbox" id="math" name="sub" value="math" />
      <label for="math">math</label>
      <input type="checkbox" id="science" name="subjects" value="sci" />
      <label for="science"> science</label>
      <input type="checkbox" id="english" name="subjects" value="eng" />
      <label for="english"> english </label>
    </div>
    <button>submit</button>
  </form>
```

→ Select element

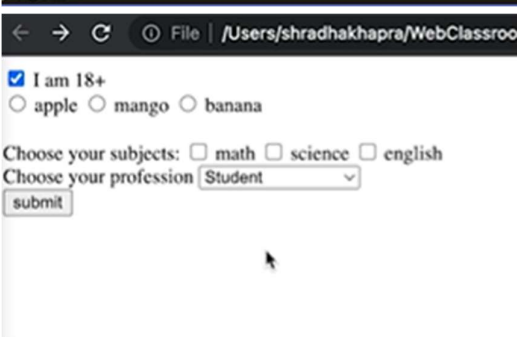
# Select Element

## Dropdown

```
<select name="profession" id="profession">  
  <option value="student">Student</option>  
  <option value="dev">Developer</option>  
</select>
```

```
</div>  
<div>  
  Choose your profession  
  <select name="profession" id="profession">  
    <option>Select your option</option>  
    <option value="stu" selected>Student</option>  
    <option value="dev">Developer</option>  
  </select>  
</div>  
<button>submit</button>  
</form>
```

→



The screenshot shows a web browser window with the address bar displaying "/Users/shradhakhapra/WebClassroom". The form contains the following elements:

- A checked checkbox labeled "I am 18+".
- Three radio buttons labeled "apple", "mango", and "banana".
- A label "Choose your subjects:" followed by three checkboxes labeled "math", "science", and "english".
- A label "Choose your profession:" followed by a dropdown menu showing "Student".
- A "submit" button.

→

## Range input element

### Input

#### Range

```
<label for="volume">Volume</label>
```

```
<input type="range" min="0" max="100" name="vol" />
```

```
</select>
</div>
<label for="vol">Select your volume level : </label>
<input type="range" id="vol" min="0" max="100" name="vol" step="10" />
<button>submit</button>
</form>
```

```
body>
</select>
</div>
<label for="vol">Select your volume level : </label>
<input
  type="range"
  id="vol"
  min="0"
  max="100"
  name="vol"
  step="10"
  value="70"
/>
<button>submit</button>
</form>
```

->if we put value = 70 then default after refresh the page also 70 will be there if we submit it will pass as 70 if we change then the value will change and passed.

reddy2096@gmail.com

## Text Area

`<label for="feedback">Please provide your valuable feedback :</label>`

`<textarea id="feedback" > </textarea>`

```
/>  
<br />  
<label for="feedback">Please give your valuable feedback : </label>  
<textarea id="feedback"></textarea>  
<button>submit</button>  
</form>
```

Choose your profession

Select your volume level :

Please give your valuable feedback :

reddy2096@gmail.com

File | /server?age=on&profession=dev&vol=70

```
<br />  
<label for="feedback">Please give your valuable feedback : </label>  
<textarea id="feedback" name="feedback"></textarea>  
<button>submit</button>
```

Select your volume level :

Please give your valuable feedback :

```
server?age=on&profession: x M <input type="range"> - HTML: x +  
File | /server?age=on&profession=dev&vol=70&feedback=it+was+good%21|
```

```
<br />  
<label for="feedback">Please give your valuable feedback : </label>  
<textarea  
  id="feedback"  
  name="feedback"  
  placeholder="write your feedback here"  
></textarea>  
<button>submit</button>
```

Select your volume level :

Please give your valuable feedback :